

Providing reliable, high-temperature space heating and domestic hot water for sm

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	a_compact_commercial_airspace_heat_pump
Industry Domain	hvac_commercial
Engine Version	PDF Engine v2.1 (Full Detail)
Target Quantity	Not specified

ARCHITECTURE AT A GLANCE

MODULES 10	BILL OF MATERIALS ROWS 0	IDENTIFIED RISKS 30
----------------------------------	--	---

ECONOMICS

ESTIMATED UNIT COST £0.00	TARGET CEILING £4,500.00	HEADROOM £4,500.00
TOTAL NON-RECURRING ENGINEERING £0.00	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING £0.00	

Cover

Page 1 of 75

Contents

1. Brief & Requirements
2. Regulatory & Compliance Posture
3. Sizing & Spatial Optimisation
4. System Modules & Architecture
5. Bill of Materials
6. Cost Waterfall & Economics
7. Risks Register
8. Supplier Shortlist & Sourcing
9. Audit Log
10. Source Attribution

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Providing reliable, high-temperature space heating and domestic hot water for small commercial premises without the risk of bringing flammable refrigerants indoors.

Mission Statement

To decarbonize UK small commercial buildings with an affordable, high-efficiency, propane-based hydronic split air-source heat pump.

Target Customers

Small to medium commercial buildings, boutique hotels, light industrial facilities, and educational premises in the UK.

Why Now (Market Timing)

Stringent F-Gas regulations are forcing a shift to low-GWP refrigerants like R290, while escalating energy costs and government net-zero targets mandate high-COP heating alternatives to gas boilers.

Full Synthesis Report

The development of a compact 30kW commercial air-source heat pump (ASHP) utilizing R290 (propane) refrigerant is a highly timely initiative for the UK small commercial market. With the UK government aggressively pursuing decarbonization through the phase-out of fossil-fuel boilers and tightening F-Gas regulations, the transition to ultra-low Global Warming Potential (GWP) refrigerants like R290 (GWP=3) is no longer optional—it is a market imperative. The specified hydronic split architecture is a critical safety and marketing advantage; by keeping the entire R290 charge strictly within the outdoor unit and circulating only water indoors, the design inherently mitigates indoor flammability and explosive risks associated with propane, significantly easing building compliance under BS EN 378. Achieving a COP of 3.5 or greater at A2/W35 utilizing a stainless steel heat exchanger and an R290-compatible scroll compressor is technically highly viable. However, reaching the £4,500 unit cost at a relatively low scale of 500 EAU will be the primary programmatic challenge. This cost target dictates a highly optimized global supply chain for heavy-duty components. Furthermore, the strict phys-

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

ical constraints (max 1200mm W x 800mm D x 1400mm H) combined with the 65dB(A) noise limit at 1 meter necessitate advanced acoustic attenuation. Dense packaging combined with compressor sound jackets, heavy-duty anti-vibration mounts, and advanced axial fan blade geometries will be required. Ultimately, achieving UK MCS certification will be the definitive hurdle to unlocking commercial viability and subsidy access in the UK.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.2 Market & Competitors

Competitor Landscape

Vaillant

Product	aroTHERM plus (Cascaded)
Pricing Strategy	£8,000 (for two 15kW units combined)
Technical Specs	Dual 15kW units cascaded for 30kW, R290 refrigerant, COP >3.6, ultra-low noise design.

STRENGTHS

Phenomenal acoustic insulation, highly trusted brand, reliable supply chain.

WEAKNESSES

Cascading two smaller units doubles the footprint and installation complexity for 30kW needs.

DIFFERENTIATION ANGLE

Providing a single, compact 30kW unit that reduces installation space, pipework complexity, and capital cost compared to cascading multiple residential models.

Clivet

Product	TiTAN R290 Commercial Range
Pricing Strategy	£6,500
Technical Specs	30-40kW monobloc commercial ASHP, double scroll compressors, R290.

STRENGTHS

Built specifically for rough commercial duty cycles, excellent BMS integration.

WEAKNESSES

Higher unit cost and slightly larger footprint that exceeds the 1200x800x1400 constraints.

DIFFERENTIATION ANGLE

Aggressive cost optimization reaching £4,500 and a strictly constrained footprint tailored to tight UK alleys and retrofitted small commercial rooftops.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Mitsubishi Electric

Product	Ecodan QHV R290
Pricing Strategy	£11,500
Technical Specs	40kW commercial monobloc, inverter-driven scroll, R290, high leaving water temperature.

STRENGTHS

World-class proprietary compressor technology, extreme reliability, high performance in deep sub-zero.

WEAKNESSES

Prohibitive pricing for smaller businesses, outputs exceed the strict 30kW sweet spot.

DIFFERENTIATION ANGLE

A much lower price point, strictly purposed for small SMEs that cannot afford large corporate capital expenditures on heating retrofits.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	£4,500.00
Max Mass	-
Target Process	
Target Material	
Tolerance Target	
Production Quantity	
Batch Size	500

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
MCS MIS 3005-I Heat Pump Standard	Regulatory Standard	2026	Dictates installation and product certification requirements for heat pumps to access UK market subsidies.
BSRIA UK Heat Pump Market Report 2023	Market Research	2026	Provides volumetric data, pricing insight, and capacity segmented demand for commercial heat pumps in the UK.
Copeland Scroll Compressors for R290 Technical Databook	Component Datasheet	2026	Critical for selecting a scroll compressor capable of outputting 30kW thermal within the COP and cost targets.
EN 14511:2022 Performance Testing	Engineering Standard	2026	Defines the exact laboratory conditions and calculation methods required to certify the A2/W35 COP >= 3.5.
UK F-Gas Regulations Update 2024	Legislation	2026	Confirms phase-down timelines for HFCs, cementing the necessity to utilize R290 (propane).

Data Sources:

[User] Founder brief text

[LLM] Gemini 3.1 Pro — research synthesis

2.1 MCS MIS 3005

Microgeneration Certification Scheme

Status: not-started

Owner: Compliance Engineer

Standard Summary

Standard for certification of heat pump products in the UK

Applicability Assessment

UK Market Entry

Engineering Design Impact

Requires proven efficiency, accurate sizing guides, and acoustic testing data

EVIDENCE REQUIRED

Accredited lab test reports, ErP label, technical files

GAP ACTION

Engage BRE or similar certified test lab for MCS type testing schedule

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.2 BS EN 378

Safety and environmental requirements for refrigerating systems

Status: not-started

Owner: Mechanical Engineer

Standard Summary

Safety governing toxic/flammable refrigerants like R290

Applicability Assessment

Outdoor Unit Design

Engineering Design Impact

Mandates ATEX-rated components near the compressor, proper ventilation, and ignition-free zones

EVIDENCE REQUIRED

Safety risk assessment (FMEA), ATEX certificates for electricals

GAP ACTION

Perform a hazard analysis on R290 leak scenarios in the outdoor casing

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.3 EN 14511

Air conditioners and heat pumps with electrically driven compressors

Status: not-started

Owner: Thermal Engineer

Standard Summary

Testing standard for rating COP and capacity

Applicability Assessment

Thermal Performance

Engineering Design Impact

Designs must be optimized specifically around the A2/W35 test points to meet the ≥ 3.5 criteria

EVIDENCE REQUIRED

Thermal lab test results

GAP ACTION

Build MVP prototype and run internal pre-compliance thermal chamber tests

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.4 UKCA/CE LVD & EMC

Electrical Equipment Safety & Electromagnetic Compatibility

Status: not-started

Owner: Electrical Engineer

Standard Summary

Mandatory electrical safety and emissions standards

Applicability Assessment

Overall System Electronics

Engineering Design Impact

Requires shielded cables, robust grounding, and compliant power filtering components

EVIDENCE REQUIRED

Declaration of Conformity, EMC/EMI test logs

GAP ACTION

Schedule EMC pre-compliance scanning at external lab

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.5 Acoustic EN 12102

Determination of sound power level

Status: not-started

Owner: NVH Engineer

Standard Summary

Standard for measuring unit noise

Applicability Assessment

Outdoor Unit Enclosure

Engineering Design Impact

Constrains axial fan speed, dictates acoustic insulation thickness to hit <65dB at 1m

EVIDENCE REQUIRED

Acoustic chamber test report

GAP ACTION

Procure noise insulating materials and perform prototype sound mapping

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION

The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND generic_room	RULES DOMAIN hvac_commercial	FEASIBLE No
---	--	-----------------------------------

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Total Internal Floor Area	25 m²
Total Internal Volume	75 m³
Available Floor Budget	21.25 m²

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
chassis_enclosure	2236 × 2236 × 2400	5	floor	mass_proportional
compressor_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
evaporator_coil	2236 × 2236 × 2400	5	floor	mass_proportional
condenser_bphe	2236 × 2236 × 2400	5	floor	mass_proportional
airflow_system	2236 × 2236 × 2400	5	floor	mass_proportional

Data Sources:

[Deterministic] Sizing solver and envelope constraints

refrigerant_circuit	2236 × 2236 × 2400	5	floor	mass_proportional
hydraulic_module	2236 × 2236 × 2400	5	floor	mass_proportional
power_inverter	2236 × 2236 × 2400	5	floor	mass_proportional
central_controller	2236 × 2236 × 2400	5	floor	mass_proportional
r290_safety_system	2236 × 2236 × 2400	5	floor	mass_proportional

Identified Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

Data Sources:

[Deterministic] Sizing solver and envelope constraints

4.1 Chassis and Enclosure Sub-assembly

Lead Time: 6 weeks

Status: draft

Module Purpose

Provides the structural backbone, weatherproofing, and primary acoustic containment for the heat pump.

Why it matters to the system

Determines the physical footprint which is critical for retrofitting into small commercial alleyways, while providing the primary defense against noise pollution and weather.

Detailed Technical Description

The chassis is the rigid superstructure of the unit, constructed primarily from galvanized and powder-coated steel to withstand extreme external weather over a 15-20 year operational lifespan. To meet the aggressive dimensions of 1200mm W x 800mm D x 1400mm H, the frame is meticulously optimized for dense internal packaging, segregating the mechanical compressor components from the airflow sections with strict compartmentalization. This compartmentalization is crucial not just for thermal efficiency, but for R290 safety to ensure leaked gases pool in ventilated, ignition-free zones.

Acoustic attenuation is heavily integrated into the physical enclosure. All external paneling features constrained-layer dampening and mass-loaded acoustic foam affixed to the interior surfaces to absorb high-frequency compressor whine and fan vibration. This assembly ensures that the entire system adheres to the strict 65dB(A) target under EN 12102.

SYSTEM INPUTS

- Ambient air
- Physical mounting anchors

SYSTEM OUTPUTS

- Structural support to internal parts
- Acoustic noise dampening
- Directed airflow path

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.1.2 Specifications & Risk Profile

Expected Key Parts

- Galvanized steel base pan
- Acoustic insulated access panels
- Heavy-duty anti-vibration rubber feet
- Internal structural bulkheads
- Powder-coated weather louvers

Identified Failure Modes

- Corrosion of base pan leading to structural collapse
- Failure of acoustic seals leading to noise limit breach
- Water ingress onto high-voltage electronics

Open Questions

- Exact thickness of acoustic foam needed to meet MCS noise criteria
- Optimal drainage path for highly intense defrost cycles

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.1.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.2 Scroll Compressor Sub-assembly

Lead Time: 14 weeks

Status: draft

Module Purpose

Compresses low-pressure R290 vapor into high-pressure vapor to drive the thermodynamic heating cycle.

Why it matters to the system

It is the prime mover of the thermodynamic cycle; its efficiency essentially dictates the system COP and overall MCS certification viability.

Detailed Technical Description

At the heart of the system is a 30kW-capable, inverter-driven scroll compressor meticulously optimized for the thermodynamic properties of R290 (propane). This module draws low-pressure, low-temperature refrigerant gas from the evaporator and mechanically compresses it, exponentially increasing its temperature and pressure before discharging it to the condenser. Because R290 operates at moderately lower discharge temperatures than legacy refrigerants but has a very high volumetric capacity, the scroll architecture is highly tuned to ensure stable operation and high isentropic efficiency even at low ambient temperatures (A2/W35 points).

Due to the stringent 65dB(A) limit, the compressor is encapsulated in a bespoke, dense multilayer acoustic jacket designed to trap mid-to-high frequency mechanical noise. Furthermore, the compressor is mounted on specialized, high-deflection elastomeric mounts that isolate structural vibrations from transmitting into the main chassis. A crankcase heater prevents liquid refrigerant migration during off-cycles.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Low-pressure R290 vapor
- 3-phase electrical power
- Inverter control signal
- High-pressure R290 vapor
- Vibrational kinetic energy
- Thermal radiation

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.2.2 Specifications & Risk Profile

Expected Key Parts

- R290 ATEX-rated Scroll Compressor
- Multi-layer acoustic dampening jacket
- Crankcase heater
- Heavy-duty vibration isolating grommets

Identified Failure Modes

- Liquid slugging destroying the internal scroll plates
- Loss of lubrication due to oil trapping in system pipework
- Insulation failure of internal motor windings causing a short circuit

Open Questions

- Actual acoustic spectrum profile under heavy frost load
- Precise lifespan of the mounting grommets under UK temperature fluctuations

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.2.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.3 Air-Side Heat Exchanger (Evaporator)

Lead Time: 12 weeks

Status: draft

Module Purpose

Absorbs latent thermal energy from ambient air to boil low-pressure liquid R290 into a gas.

Why it matters to the system

The surface area and efficiency of this module are directly responsible for the heat pump's ability to extract free energy from the ambient air, making the COP > 3.5 target possible.

Detailed Technical Description

The evaporator module is a massive, multi-row fin-and-tube heat exchanger folded into either an L or V configuration to maximize surface area within the constrained 1200x800mm footprint. Liquid R290 exiting the expansion valve flashes into the copper tubing grid, rapidly boiling as it absorbs thermal energy harvested from the outdoor air being pulled through the aluminum fins. The fins are coated with a hydrophilic layer to expedite the shedding of condensate water, significantly delaying the formation of frost during typical cold, damp UK winters.

Because maintaining a high COP inherently requires maximizing evaporator efficiency, the tubes feature internal micro-grooving to increase turbulence and wetted surface area for the refrigerant. The sub-assembly also heavily integrates condensate management; an active drip tray with localized heating strips ensures that during reverse-cycle defrosts, the shedding ice rapidly sublimates and drains clear of the unit base without re-freezing and blocking airflow.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Cold ambient air flow - Low-pressure R290 liquid/vapor mixture	- Chilled ambient air out - Low-pressure R290 superheated vapor - Condensate water (during frost/defrost)

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.3.2 Specifications & Risk Profile

Expected Key Parts

- Hydrophilic coated aluminum fins
- Internally grooved copper tubing
- Defrost loop/heater element
- Defrost temperature thermistor
- Drainage drip tray

Identified Failure Modes

- Ice buildup distorting and crushing aluminum fins
- Micro-leaks at copper return bends due to thermal cycling fatigue
- Corrosion of copper tubing in industrial or coastal atmospheres

Open Questions

- Defrost cycle optimal frequency in severe UK damp/cold mix
- Exact pressure drop across the thick fin matrix

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.3.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.4 Water-Side Heat Exchanger (Condenser)

Lead Time: 10 weeks

Status: draft

Module Purpose

Transfers heat from the hot high-pressure R290 gas directly into the building's hydronic heating circuit.

Why it matters to the system

This sub-assembly safely acts as the thermal bridge between the volatile R290 outdoor system and the safe indoor water circuit.

Detailed Technical Description

The condenser in this hydronic split architecture is a highly compact, counter-flow Brazed Plate Heat Exchanger (BPHE). High temperature, high pressure R290 discharged from the compressor enters one side of the plate matrix, while return water from the indoor heating loop is circulated through the opposing micro-channels. The extreme thermal conductivity of the stainless steel corrugated plates allows a massive 30kW of thermal energy transfer within a relatively tiny footprint. As the R290 rejects its heat, it undergoes a phase change back into a subcooled liquid.

Given the performance dictates of EN 14511 (A2/W35), the BPHE is engineered with tight approach temperatures in mind, ensuring the smallest possible delta between condensing temperature and leaving water temperature. The entire sub-assembly is heavily wrapped in closed-cell elastomeric insulation to eliminate parasitic thermal loss to the outside air and to prevent external condensation during summertime cooling modes, if applicable.

SYSTEM INPUTS

- Hot high-pressure R290 gas
- Return water from building (e.g., 30°C)

SYSTEM OUTPUTS

- Subcooled high-pressure R290 liquid
- Supply water to building (e.g., 35°C+)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.2 Specifications & Risk Profile

Expected Key Parts

- AISI 316 Stainless Steel Brazed Plate Heat Exchanger
- Vapor-proof elastomeric thermal insulation
- Water temperature thermistors (flow/return)
- Mounting brackets

Identified Failure Modes

- Internal plate rupture mixing water and R290
- Fouling/scaling on the water side dropping heat transfer efficiency
- Freezing of the water channel during extreme winter off-cycles resulting in catastrophic burst

Open Questions

- Vulnerability to poor indoor water quality (scale/sludge)
- Exact required number of plates to hit COP 3.5 given cost strictly constrained to £4500

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.5 Active Airflow System

Lead Time: 12 weeks

Status: draft

Module Purpose

Pulls tremendous volumes of ambient air through the evaporator coil while maintaining strictly constrained noise levels.

Why it matters to the system

Controls the delicate balance between required high-volume thermal extraction and the strict UK noise regulations for commercial zoning.

Detailed Technical Description

Due to the 1400mm height constraint and 30kW capacity requirement, a single massive fan is impractical. Therefore, this module comprises a stacked twin-fan array utilizing advanced Electronically Commutated (EC) axial fans. These highly efficient 630mm fans create the low-pressure zone necessary to pull thousands of cubic meters of ambient air per hour through the high-resistance evaporator coil. The variable speed EC motors respond to PWM signals from the main controller, allowing them to spin down during low thermal load to save energy and emit near-zero noise.

Acoustics dictate the specific design of this sub-assembly. To satisfy the <65dB(A) limit at 1 meter dictated by EN 12102, the fan blades employ biomimetic, swept trailing edges (owl-wing) to dramatically reduce turbulent vortex shedding (the primary source of broadband fan swoosh). Additionally, the fans are mounted within deep aero-acoustic cowlings designed to smooth the exhaust jet and reduce blade-pass frequency noise propagation.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Ambient air - Single-phase electrical power - PWM Control signal	- High-volume directional air exhaust - Acoustic noise

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.2 Specifications & Risk Profile

Expected Key Parts

- Twin stacked 630mm EC axial fans
- Aerodynamic swept owl-wing fan blades
- Noise isolating fan cowlings
- Bird/debris wire grille guards

Identified Failure Modes

- EC motor bearing failure leading to seizure and thermal cutout
- Ice throw from blades causing physical imbalance and shattering
- Resonant vibration coupling with the main chassis at specific RPMs

Open Questions

- Real-world pressure drop of the finalized chassis louvers
- Acoustic interaction effects between the upper and lower fan wakes

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.6 Refrigerant Control Circuit

Lead Time: 10 weeks

Status: draft

Module Purpose

Regulates phase changes, flow rates, safety limits, and directional cycling of the R290 refrigerant.

Why it matters to the system

Precision metering of R290 determines real-world COP; inaccurate expansion leads to wasted energy or compressor destruction.

Detailed Technical Description

This physical network of copper pipework and electromechanical brass components functions as the 'nervous system' for the thermodynamic cycle. High-pressure liquid R290 leaving the condenser BPHE passes through a filter drier to remove trace moisture and particulates before reaching the Electronic Expansion Valve (EXV). The EXV acts as a precise throttle; as the liquid is forced through this minuscule orifice, it rapidly drops in pressure and temperature, entering the evaporator as a highly reactive liquid/vapor spray.

Crucially, this module contains the 4-way reversing valve which allows the heat pump to literally operate backwards. By redirecting hot discharge gas from the compressor straight into the outdoor evaporator coil, it actively melts away accumulated frost. Accumulators and receiver tanks are plumbed into the lines to store excess R290 charge during different operating states, ensuring the compressor is neither starved of gas nor flooded with incompressible liquid.

SYSTEM INPUTS

- Subcooled R290 liquid
- Control signals

SYSTEM OUTPUTS

- Metered 2-phase R290 spray
- Reversed flow (during defrost)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.6.2 Specifications & Risk Profile

Expected Key Parts

- Electronic Expansion Valve (EXV)
- 4-way reversing valve
- Liquid receiver and suction accumulator
- High/Low pressure transducers
- Filter drier

Identified Failure Modes

- Stepper motor failure on the EXV causing flow starvation
- 4-way valve sticking mid-stroke causing cycle collapse
- Debris blocking the expansion orifice

Open Questions

- Optimal pipe routing to avoid vibration fracturing
- Exact internal charge volume optimization for the receiver

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.6.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.7 Hydronic Circulation Module

Lead Time: 8 weeks

Status: draft

Module Purpose

Circulates heated water from the condenser BPHE to the building's indoor heating loop safely and efficiently.

Why it matters to the system

Simplifies installation for commercial tradesmen while protecting the heat exchanger from instantaneous freezing or boiling faults.

Detailed Technical Description

This sub-assembly bridges the heat pump to the building's actual heating infrastructure. At its core is a high-efficiency variable-speed circulation pump engineered to maintain correct delta-T across the condenser. By adjusting the water flow rate, the pump ensures that heat produced by the thermodynamic cycle is carried away efficiently; if flow stops while the compressor is running, catastrophic high-pressure faults occur within seconds.

To ensure straightforward BS EN 378 and plumbing compliance, this module integrates critical hydronic safety elements directly inside the outdoor unit. This includes an expansion vessel to handle water volume changes during heating, an automatic air purger to eliminate cavitating oxygen from the lines, and a flow switch that acts as a hard safety interlock against the compressor. All plumbing connections emerge at the rear of the unit as standard threaded stainless or brass fittings for easy UK installer integration.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Heated water from BPHE - Mains electrical power - Return water from building	Pressurized hot water flow to building

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.7.2 Specifications & Risk Profile

Expected Key Parts

- Variable speed circulation pump
- Expansion vessel
- Flow switch / meter
- Pressure relief safety valve (PRV)
- Air purger

Identified Failure Modes

- Pump impeller jammed by sludge/magnetite from old building pipes
- Flow switch sticking 'closed', masking a dry-fire scenario
- PRV leaking precious system pressure over time

Open Questions

- Head pressure capabilities needed for average multi-story business premises
- Reliability of flow switches in high-glycol mixtures

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.7.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.8 Power and Inverter Drive Module

Lead Time: 16 weeks

Status: draft

Module Purpose

Rectifies, filters, and conditionally converts UK 3-phase mains to drive variable speed high-power motors.

Why it matters to the system

Inverter technology drives the primary energy savings and COP gains over legacy fixed-speed AC machinery while satisfying commercial grid limits.

Detailed Technical Description

The heavy-duty electrical heart of the heatpump, the Power and Inverter Drive Module receives standard 400V 3-phase grid power and transforms it for advanced appliance usage. The primary component is the high-power Variable Frequency Drive (VFD), which rectifies AC into DC, then fires IGBTs (Insulated-Gate Bipolar Transistors) to synthesize an optimized sinusoidal AC waveform. This completely variates the speed of the scroll compressor, allowing the heat pump to modulate smoothly from 10kW up to 30kW capacity rather than brute-force cycling on and off. This is critical for achieving an ErP A+++ rating and achieving the MCS certification targets.

Because switching 30kW creates massive electromagnetic interference (EMI), this module is densely packed with heavy toroidal line chokes and capacitors to guarantee UKCA/CE EMC compliance. Furthermore, the inverter drive generates substantial waste heat; therefore, its aluminum finned heatsink is strategically embedded directly into the airflow path of the main EC fans, simultaneously cooling the electronics while reclaiming trace heat into the system.

SYSTEM INPUTS	SYSTEM OUTPUTS
400V 3-phase 50Hz AC mains	- DC variable voltage/frequency for compressor - DC/AC lower voltage feeds for fans, heaters 5V/12V DC for logic boards

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.8.2 Specifications & Risk Profile

Expected Key Parts

- 30kW 3-phase Inverter Drive (VFD)
- EMC line filter chokes
- Giant DC-link capacitors
- Solid State Relays (SSRs)
- Aluminum heatsink

Identified Failure Modes

- IGBT burnout due to inadequate heatsink cooling
- DC-link capacitor electrolyte venting due to overvoltage spikes
- EMC emissions interfering with the building's Wi-Fi or BMS systems

Open Questions

- Extent of EMI shielding required for CE verification
- Heat load rejection limits on the custom inverter heat sink

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.8.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.9 Central Control Platform

Lead Time: 12 weeks

Status: draft

Module Purpose

The main brain executing thermodynamic cycles, safety algorithms, and external BMS communication.

Why it matters to the system

Hardware is only as good as its software; thermodynamic optimization and frost handling logic is governed exclusively here.

Detailed Technical Description

Operating as the system orchestrator, the Central Control Platform processes over 40 distinct sensor inputs per second to master the complex refrigeration cycle. Using highly proprietary PID loops, it continuously adjusts the compressor inverter speed, EXV stepper position, and fan rpm to chase tight leaving-water temperature setpoints while optimizing COP dynamically. During deep frost conditions, the controller utilizes predictive algorithms comparing ambient capability against suction pressure degradation to trigger the specialized 4-way valve reverse defrost cycles seamlessly.

In the small commercial market, standalone thermostats are rarely sufficient. This board features extensive native protocol support (Modbus RTU, BACnet) to interface with established commercial Building Management Systems (BMS). It logs critical operation hours, error histories (vital for warranty mitigation), and energy consumption parameters required for MCS MIS 3005-I validation.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Thermistor data (air, gas, water) - Pressure transducer signals - User Interface / BMS commands	- PWM logic to inverter and fans - Stepper signals to EXV - Modbus / BACnet outputs

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.9.2 Specifications & Risk Profile

Expected Key Parts

- Main microcontroller PCB (e.g., Carel)
- I/O terminal block array
- EEPROM parameter chip
- Modbus communication gateway

Identified Failure Modes

- Firmware corruption locking out operation
- Micro-arch flash logic failure resulting in bad EXV driving (system flood-back)
- I/O port destruction from lightning-induced ground surges

Open Questions

- Optimal PID tuning parameters for diverse UK weather conditions
- Total lines of code necessary to comply with safety interlock latency rules

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.9.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.10 R290 Safety and Diagnostics Element

Lead Time: 8 weeks

Status: draft

Module Purpose

Monitors the chassis for highly flammable gas leaks and actively prevents ATEX ignition scenarios.

Why it matters to the system

Absolute prerequisite for legal market entry; failures regarding flammable A3 refrigerants lead to catastrophic loss of life and extreme legal liability.

Detailed Technical Description

Utilizing ~3kg of R290 (propane) invokes strict mandates under BS EN 378 due to its A3 extreme flammability rating. Because propane is heavier than air, any micro-leak from the pressurized refrigeration line will 'pool' in the bottom of the enclosure, creating an explosive atmosphere waiting for an electrical arc. The Safety and Diagnostics Element is an independent, highly resilient network of ATEX-rated components installed at the chassis floor to mitigate this risk completely.

Two continuously monitoring catalytic or infrared gas detectors constantly sample the internal air space. In the event 15% of the Lower Flammability Limit (LFL) is reached, this sub-assembly triggers an independent hardware relay. This relay instantly severs power to the compressor and main board, while simultaneously firing a specialized spark-proof purge fan to forcibly exhaust the pooled propane into the open atmosphere, nullifying the explosion risk. All contactors and switches inside this zone are potted into hermetically sealed explosion-proof housings.

SYSTEM INPUTS

- Internal ambient air sampling
- Dedicated power feed

SYSTEM OUTPUTS

- Emergency shutdown signal
- Purge fan activation

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.10.2 Specifications & Risk Profile

Expected Key Parts

- Infrared/catalytic R290 gas sensors
- ATEX-rated electrical hermetic junctions
- Dedicated ATEX purge exhaust fan
- Hardwired safety relay (independent of main PCB)

Identified Failure Modes

- Gas sensor drift causing false positive shut-downs
- Purge fan failure leaving explosive gas trapped inside unit
- Relay welded shut failing to kill main power during leak event

Open Questions

- Lifespan of infrared gas sensors in high-vibration environments
- Airflow draft vectors and optimal sensor placement within crowded casing

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.10.3 Cost Breakdown

No bill of materials rows allocated to this module yet.

Data Sources:

[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

5. Bill of Materials

No parts in bill of materials.

6. Cost Waterfall & Economics

UNIT COST £0.00	TARGET CEILING £4,500.00	HEADROOM £4,500.00
--------------------	-----------------------------	-----------------------

Make vs Buy Split

PURCHASED (BUY) TOTAL £0.00	CUSTOM (MAKE) TOTAL £0.00	MAKE % 0%
--------------------------------	------------------------------	--------------

Per-Module Roll-up

Per-module cost data not available.

Non-recurring engineering and overheads

Total non-recurring engineering (Tooling, £0.00
Safety Testing, Compliance)

Grand Total (Unit + Non-recurring Engineering)	£0.00
--	-------

Data Sources: [Deterministic] Domain overhead multiplier model

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

Data Sources:

[Deterministic] Domain overhead multiplier model

7.1 Chassis and Enclosure Sub-assembly Risks

Risk 1: Resonant frequency amplification during peak compressor load		Base Score: S:4 × L:3 = 12
EXISTING CONTROLS	PLANNED MITIGATION	
None	Perform FEA modal analysis on base frame and add cross-bracing	
Owner: Mechanical Engineer		

Risk 2: Corrosion in coastal environments		Base Score: S:3 × L:2 = 6
EXISTING CONTROLS	PLANNED MITIGATION	
None	Upgrade powder coat specification to C4 standard	
Owner: Manufacturing Engineer		

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.1 Chassis and Enclosure Sub-assembly Risks

Risk 3: Panel alignment tolerance stacks up, blocking access

Base Score: S:2 × L:4 = 8

EXISTING CONTROLS

None

PLANNED MITIGATION

Tighten CNC folding tolerances and use slot/tab integration

Owner: Mechanical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Scroll Compressor Sub-assembly Risks

Risk 4: Acoustic limit breach at full operational RPM

Base Score: $S:5 \times L:3 = 15$

EXISTING CONTROLS

None

PLANNED MITIGATION

Cap inverter maximum frequency in software based on ambient noise mode

Owner: NVH Engineer

Risk 5: Oil wash-out during low-capacity modulation

Base Score: $S:4 \times L:2 = 8$

EXISTING CONTROLS

None

PLANNED MITIGATION

Program oil return cycles into the primary control logic

Owner: Thermodynamics Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Scroll Compressor Sub-assembly Risks

Risk 6: Supply chain bottleneck for specialized R290 compressor		Base Score: S:4 × L:4 = 16
EXISTING CONTROLS		PLANNED MITIGATION
None		Dual source between Copeland and Danfoss units
Owner: Supply Chain Manager		

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Air-Side Heat Exchanger (Evaporator) Risks

Risk 7: Excessive frosting leading to airflow choke

Base Score: S:4 × L:4 = 16

EXISTING CONTROLS

None

PLANNED MITIGATION

Increase fin spacing on outer rows and implement predictive defrost logic

Owner: Thermal Engineer

Risk 8: R290 leak at brazed U-bends

Base Score: S:5 × L:2 = 10

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement 100% helium leak test at factory EOL

Owner: Quality Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Air-Side Heat Exchanger (Evaporator) Risks

Risk 9: Drip tray overflow damaging base enclosure

Base Score: S:3 × L:3 = 9

EXISTING CONTROLS

None

PLANNED MITIGATION

Widen drainage channel and specify a high-power trace heater

Owner: Mechanical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Water-Side Heat Exchanger (Condenser) Risks

Risk 10: Freezing burst of BPHE during power outage in sub-zero weather Base Score: S:5 × L:3 = 15	
EXISTING CONTROLS	PLANNED MITIGATION
None	Recommend anti-freeze valve in installation manual and add trace heating
Owner: Compliance Engineer	

Risk 11: Underperformance of heat transfer due to poor flow distribution Base Score: S:3 × L:2 = 6	
EXISTING CONTROLS	PLANNED MITIGATION
None	Specify proprietary asymmetric plate patterns from Sweb/Alfa Laval
Owner: Thermal Engineer	

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Water-Side Heat Exchanger (Condenser) Risks

Risk 12: Cost overrun for stainless material

Base Score: S:3 × L:4 = 12

EXISTING CONTROLS

None

PLANNED MITIGATION

Negotiate long-term volume agreements based on 500 EAU

Owner: Supply Chain Manager

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.5 Active Airflow System Risks

Risk 13: Exceeding 65dB(A) limit at full defrost speed

Base Score: S:5 × L:3 = 15

EXISTING CONTROLS

None

PLANNED MITIGATION

Work with ebm-papst on bespoke blade geometries and implement night-mode decibel limits

Owner: NVH Engineer

Risk 14: Ingress of debris/leaves blocking EC motor cooling

Base Score: S:3 × L:4 = 12

EXISTING CONTROLS

None

PLANNED MITIGATION

Design specific finned grille configuration that maintains aero-efficiency but blocks debris

Owner: Mechanical Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.5 Active Airflow System Risks

Risk 15: Fan blade imbalance out of the box

Base Score: S:2 × L:2 = 4

EXISTING CONTROLS

None

PLANNED MITIGATION

100% factory dynamic balancing requirement on EOL test

Owner: Quality Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Refrigerant Control Circuit Risks

Risk 16: Vibration fatigue causing pipe rupture and total R290 loss Base Score: S:5 × L:3 = 15	
EXISTING CONTROLS	PLANNED MITIGATION
None	Introduce specialized vibration absorber tubings and 3D map line stress
Owner: Mechanical Engineer	

Risk 17: Moisture in the R290 charge causing acid formation Base Score: S:4 × L:2 = 8	
EXISTING CONTROLS	PLANNED MITIGATION
None	Mandate high-vacuum purge procedures during factory charge and spec oversized filter drier
Owner: Manufacturing Engineer	

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Refrigerant Control Circuit Risks

Risk 18: 4-way valve fails to shift dynamically in deep freezing temps Base Score: S:3 × L:3 = 9	
EXISTING CONTROLS	PLANNED MITIGATION
None	Ensure pressure differential is verified before shifting valve signal
Owner: Thermodynamics Engineer	

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 Hydronic Circulation Module Risks

Risk 19: Dry running the BPHE due to trapped air loop		Base Score: S:4 × L:3 = 12
EXISTING CONTROLS		PLANNED MITIGATION
None		Add secondary low-pressure electronic transducer interlock on the water side
Owner: Controls Engineer		

Risk 20: Grit destruction of the modern EC pump		Base Score: S:3 × L:4 = 12
EXISTING CONTROLS		PLANNED MITIGATION
None		Mandate external magnetic filter installation in warranty terms
Owner: Compliance Engineer		

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 Hydronic Circulation Module Risks

Risk 21: Expansion vessel membrane rupture

Base Score: S:2 × L:2 = 4

EXISTING CONTROLS

None

PLANNED MITIGATION

Over-spec vessel capacity and utilize EPDM membranes

Owner: Mechanical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.8 Power and Inverter Drive Module Risks

Risk 22: Failure of EMC pre-compliance testing

Base Score: S:4 × L:3 = 12

EXISTING CONTROLS

None

PLANNED MITIGATION

Engage expert EMC consultant for board layout and incorporate multi-stage line filters

Owner: Electrical Engineer

Risk 23: Harmonic distortion fed back into the commercial electrical grid

Base Score: S:4 × L:2 = 8

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement Active Power Factor Correction (PFC) circuitry

Owner: Electrical Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.8 Power and Inverter Drive Module Risks

Risk 24: Inverter shutdown off high ambient temps (summer water heating)

Base Score: S:3 × L:3 = 9

EXISTING CONTROLS

None

PLANNED MITIGATION

Provide dedicated cooling draft path in chassis for electrical box

Owner: Mechanical Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Central Control Platform Risks

Risk 25: Software bug causes 4-way valve flip while compressor running at full speed Base Score: S:5 × L:2 = 10	
EXISTING CONTROLS	PLANNED MITIGATION
None	Hardcode mechanical delay interlock timers at firmware root level
Owner: Controls Engineer	

Risk 26: BMS integration failure on client site Base Score: S:2 × L:4 = 8	
EXISTING CONTROLS	PLANNED MITIGATION
None	Pre-certify Modbus registers and provide extensive API documentation
Owner: Software Engineer	

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Central Control Platform Risks

Risk 27: Component shortage delaying PCB fab

Base Score: S:4 × L:3 = 12

EXISTING CONTROLS

None

PLANNED MITIGATION

Partner with Tier 1 vendor like Carel for guaranteed supply chains

Owner: Supply Chain Manager

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.10 R290 Safety and Diagnostics Element Risks

Risk 28: Ignition of leaked R290 inside compressor zone

Base Score: S:5 × L:1 = 5

EXISTING CONTROLS

None

PLANNED MITIGATION

Complete segregation of low-voltage electricals, ATEX rating for all components in danger zone

Owner: Compliance Engineer

Risk 29: False positive gas alarms leading to continuous commercial downtime

Base Score: S:3 × L:4 = 12

EXISTING CONTROLS

None

PLANNED MITIGATION

Use high-quality IR dual-wavelength sensors mapped to compensate for humidity

Owner: Electrical Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.10 R290 Safety and Diagnostics Element Risks

Risk 30: Fail-safe relay circuit fails to trigger purge fan

Base Score: S:5 × L:2 = 10

EXISTING CONTROLS

None

PLANNED MITIGATION

Design circuit as normally-closed so total power loss defaults to safe logic state

Owner: Mechanical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

8. Supplier Shortlist

No supplier matches found during the search phase.

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	a_compact_commercial_airsource_heat_pump
Research Result	Generated
System Modules	10 Modules
Bill of Materials Lines	0 Parts
Cost Analysis	Pending
Specialist Reviews	0 Completed
Supplier Shortlists	0 Matched

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition